

Retrieval-Augmented Generation (RAG)

Retrieval-augmented generation (RAG) is a technique that enables large language models (LLMs) to retrieve and incorporate new information from external data sources. With RAG, LLMs first refer to a specified set of documents, then respond to user queries. These documents supplement information from the LLM's pre-existing training data. This allows LLMs to use domain-specific and/or updated information that is not available in the training data. For example, this enables LLM-based chatbots to access internal company data or generate responses based on authoritative sources. The technique was first proposed in 2020 and has since become a widely adopted approach in modern AI systems.

RAG improves LLMs by incorporating information retrieval before generating responses. Unlike LLMs that rely on static training data, RAG pulls relevant text from databases, uploaded documents, or web sources. According to Ars Technica, "RAG is a way of improving LLM performance, in essence by blending the LLM process with a web search or other document look-up process to help LLMs stick to the facts." This method helps reduce AI hallucinations, which have caused chatbots to describe policies that don't exist, or recommend nonexistent legal cases to lawyers that are looking for citations to support their arguments.

RAG also reduces the need to retrain LLMs with new data, saving on computational and financial costs. Beyond efficiency gains, RAG also allows LLMs to include sources in their responses, so users can verify the cited sources. This provides greater transparency, as users can cross-check retrieved content to ensure accuracy and relevance.

The term retrieval-augmented generation (RAG) was introduced in a 2020 paper that described combining a parametric language model with a non-parametric external memory accessed through retrieval at inference time.

== RAG and LLM limitations ==

LLMs can provide incorrect information. For example, when Google first demonstrated its LLM tool "Google Bard" (later re-branded to Gemini), the LLM provided incorrect information about the James Webb Space Telescope. This error contributed to a \$100 billion decline in Google's stock value.

RAG is used to prevent these errors, but it does not solve all the problems. For example, LLMs can generate misinformation even when pulling from factually correct sources if they misinterpret the context. MIT Technology Review gives the example of an AI-generated response stating, "The United States has had one Muslim president, Barack Hussein Obama." The model retrieved this from the rhetorical chapter title "Barack Hussein Obama: America's First Muslim President?" in the book *Faith in the New Millennium: The Future of Religion and American Politics*. The LLM did not "know" or "understand" the context of the title, generating a false statement.

LLMs with RAG are programmed to prioritize new information. This technique has been called "prompt stuffing." Without prompt stuffing, the LLM's input is generated by a user; with prompt stuffing, additional relevant context is added to this input to guide the model's response. This approach provides the LLM with key information early in the prompt, encouraging it to prioritize the supplied data over pre-existing training knowledge.

== Process ==

Retrieval-augmented generation (RAG) enhances large language models (LLMs) by incorporating an information-retrieval mechanism that allows models to access and utilize additional data beyond their original training set. Ars Technica notes that "when new information becomes available, rather than having to retrain the model, all that's needed is to augment the model's external knowledge base with the updated information" ("augmentation"). IBM states that "in the generative phase, the LLM draws from the augmented prompt and its internal representation of its training data to synthesize" an answer.

=== RAG key stages ===

Typically, the data to be referenced is converted into LLM embeddings, numerical representations in the form of a large vector space. RAG can be used on unstructured (usually text), semi-structured, or structured data (for example knowledge graphs). These embeddings are then stored in a vector database to allow for document retrieval.

Given a user query, a document retriever is first called to select the most relevant documents that will be used to augment the query. This comparison can be done using a variety of methods, which depend in part on the type of indexing used.

The model feeds this relevant retrieved information into the LLM via prompt engineering of the user's original query. Newer implementations (as of 2023) can also incorporate specific augmentation modules with abilities such as expanding queries into multiple domains and using memory and self-improvement to learn from previous retrievals.

Finally, the LLM can generate output based on both the query and the retrieved documents. Some models incorporate extra steps to improve output, such as the